CHEMISTRY IN THE SERVICE OF MANKIND

Polymers

Polymers are defined as high molecular mass and long-chain molecules made by joining many small molecules called monomers. Polymers are also called macromolecules.

Polymerization

The process by which polymers are formed from their corresponding monomers is called polymerization.

$$\text{Monomers} \xrightarrow{\text{Polymerization}} \text{Polymers}$$

Types of polymerization

1. Addition polymerization

It is the formation of polymers by repeated addition of monomers.
For eg: When a large number of ethane molecules polymerize to form polythene

2. Condensation polymerization

It is the process of polymer formation by combining two types of monomers with the elimination of small molecules like H_2O, NH_3, HCl, etc.
For eg: The formation of Nylon-6,6 from adipic Acid and hexamethylene diamine

Classification of Polymers

A. Classification based on type polymerization

1) Addition polymer

The polymers formed by adding monomer units without eliminating any molecule are called addition polymers. Monomer units are usually unsaturated compounds.

Examples:

- Polyethene: polymer of ethene monomer
- Polyvinyl Chloride (PVC): polymer of vinyl chloride monomer
- Polystyrene: polymer of styrene monomer
- Teflon: polymer of tetrafluoroethene monomer
- Benzene: polymer of ethyne (acetylene) monomer

2) Condensation polymer/ step growth polymer

The polymers formed by addition of monomer units with the elimination of smaller molecules are called condensation polymers. Monomer units are usually two or more with different functional groups.

Example:

- Nylon-6,6: polymer of hexamethylene diamine and adipic acid monomer
- Bakelite: polymer of phenol and formaldehyde monomer
- Polyester: polymer of terephthalic and ethylene glycol monomer

B. Classification based on molecular forces

1) Elastomers

The polymers in which chains are held together by very weak intermolecular forces are called elastomers.

Examples: natural rubber, synthetic rubber, silicone rubber, etc.

2) Fibres

The polymers having the strongest intermolecular force of attraction are called fibres. They have high tensile strength and minimum elasticity.

Example: Polyester, orlon (Polyacrylonitrile-PAN), terylene, etc.

C. Classification based on similarities of monomers/ monomer unit

1) Homopolymers

The polymers formed by only one type of monomer are called homopolymers. The polymerization of a single monomer to form polymer is called homopolymerizarion.

Examples of homopolymer:

- Natural rubber: polymer of isoprene monomer
- Polythene: polymer of ethene monomer
- Polyvinyl Chloride - polymer of vinyl chloride monomer

- Polystyrene: polymer of styrene (2-phenyl ethene) monomer
- Polytetrafluoroethane [PTFE or Teflon) – polymer of tetrafluoroethene

2) Copolymers

The polymers formed by more than one type of monomer unit are called copolymers. Polymerization of more than one type of monomer is called copolymerization.
Examples of copolymer:

- Nylon-6,6: polymer of hexamethylene diamine and adipic acid monomer
- Bakelite: polymer of phenol and formaldehyde monomer

D. Classification based on origin

1) Natural Polymer

The polymers formed in the body of plants and animals are natural polymer
Examples:

- Protein: polymer of amino acid monomer
- Nucleic acid: DNA/RNA are polymers of nucleotide/ ribonucleotide monomer
- Natural rubber: polymer of isoprene monomer

2) Synthetic polymer

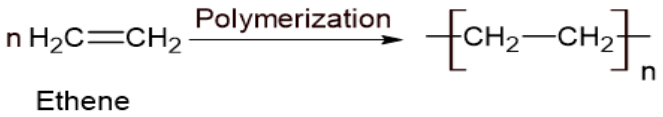
The polymers synthesized/ prepared in lab or chemical industries are called synthetic polymers.
Examples:

- Polyethene: polymer of ethene monomer
- PVC: polymer of vinyl chloride monomer
- Polystyrene polymer of styrene
- Nylon-6,6: polymer of hexamethylene diamine and adipic acid
- Bakelite: polymer of phenol and formaldehyde

Some Synthetic Polymers

1. Polythene

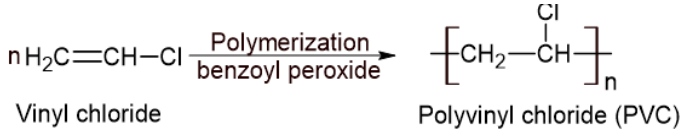
It is a homopolymer and addition polymer obtained by polymerizing ethene under high temperature and pressure. The monomer of polythene is ethene.



Uses: packaging, making bottles, buckets and electrical insulation.

2. PVC (Polyvinyl chloride)

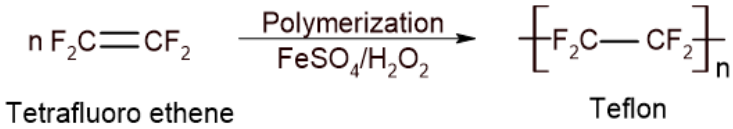
It is a homopolymer and addition polymer obtained by polymerizing vinyl chloride under high pressure and temperature. The monomer of PVC is vinyl chloride.



Uses: making plastic pipe, floor tile, raincoats, blood bags, etc.

3. Teflon (PTFE or Fluon)

It is a homopolymer and addition polymer obtained by polymerizing tetrafluoro ethene at high pressure and temperature. The monomer of Teflon is tetrafluoro ethene.



Uses: It is tough and resistant to heat so it is used for coating utensils, gaskets, valves, etc.

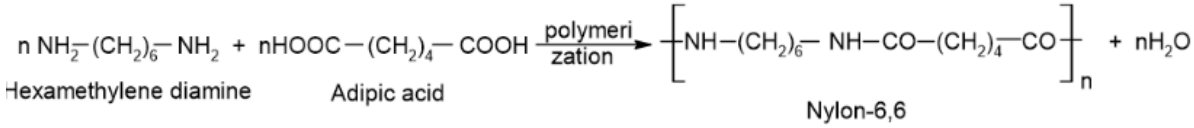
4. Polystyrene

It is a homopolymer, thermoplastic and addition polymer obtained by polymerizing styrene. The monomer of polystyrene is styrene.

Uses: It is used in the manufacture of TV, radio, refrigerator, plastic cups etc.

5. Nylon-6,6

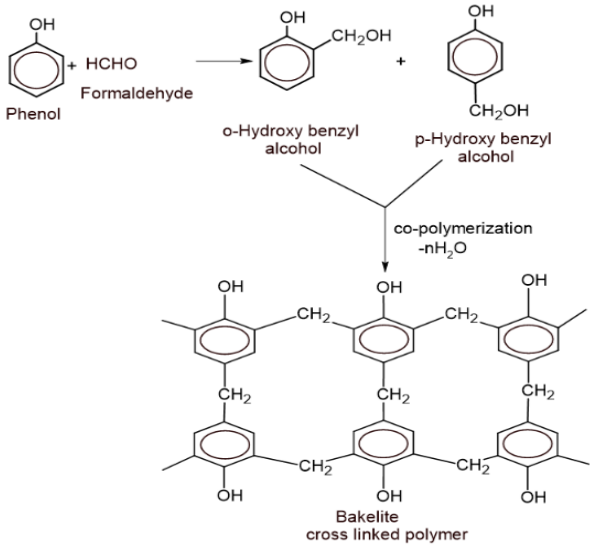
It is a copolymer and condensation polymer obtained by polymerizing adipic acid and hexamethylene diamine.



Uses: It is used to make fabrics, carpets, and ropes in textile industries.

6. Bakelite

It is a copolymer and condensation polymer obtained by polymerizing phenol and formaldehyde. The monomers of bakelite are phenol and formaldehyde.



Uses: for making handles on electrical and cooking utensils, combs, fountain pens, etc. Novolac is used in paints.

Dyes

The natural synthetic colored substances used to impart color to textiles, foodstuffs, leather, etc. are called dyes.

Characteristics of Dyes

- It should have suitable color.
- It should be able to fix itself.
- It should be fast to light.
- It should be resistant to water, dil. acids, alkalis, and various organic solvents.

Structure of dyes

a. Chromophores

The unsaturated group such a –N=N-, -0-N=0, >C=0, etc which are responsible for the color of dye are called chromophores.

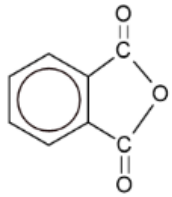
b. Auxochromes

The groups which increase the effect of chromophores and help in the attachment of dyes to other materials are called auxochromes. Eg: -OH, $-NH_2$, -NHR, etc.

Classification of Dyes by Structure

They possess a specific structural unit in them.

1. Phthalein dye



Structural unit	Example	Use
	phenolphthalein	indicator in acid-base titration

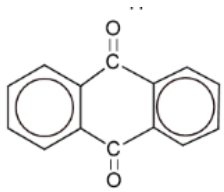
2. Azo dye

Structural unit	Example	Use
-N=N-	p-amino azobenzene	dye for oil

3. Nitro dye

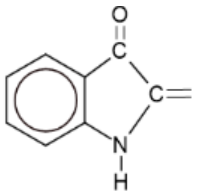
Structural unit	Example	Use
$-NO_2$	picric acid, Martius yellow, etc.	dye for wool

4. Anthraquinone dye



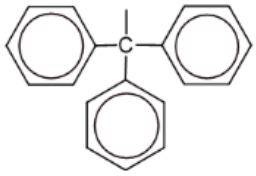
Structural unit	Example	Use
	alizarin	dyeing cotton and printing inks

5. Indigo dye



Structural unit	Example	Use
	indigo	dye cotton, wool

6. Triphenyl methane



Structural unit	Example	Use
	Crystal violet	dyeing acrylic fibre, leather, etc.

Classification of Dyes by Methods of Application

1. Direct dyes

- ✓ Used for dyeing fabrics
- ✓ Water soluble
- ✓ Example: Martius yellow

2. Vat dyes

- ✓ Used for dyeing cotton and cellulose fibres
- ✓ Water insoluble
- ✓ Example: Indigo

3. Mordant dyes

- ✓ Used for dyeing wool, cotton and nylon
- ✓ Example: Alizarin

4. Azoic dyes (Ingrain dyes)

- ✓ Used for dyeing cotton, cellulose fibre and nylon
- ✓ Water insoluble azo dyes

5. Disperse dyes

- ✓ Used for synthetic fibres like Orlon, nylon, polyester, etc.
- ✓ Also called synthetic fibre dyes

- ✓ Water insoluble
- ✓ Example: 1-amino-4-hydroxy anthraquinone

Classification of Dyes on the basis of origin

1. Natural Dyes

The dyes derived from plant sources like roots, barks, leaves, fungi, lichens, etc. are called natural dyes. Example: Indigo-derived from leaves of *Indigofera* species.

2. Synthetic Dyes

The dyes prepared in labs or industries are called synthetic dyes. Example: phenolphthalein, p-hydroxyazobenzene, p-aminoazobenzene, picric acid, methyl orange, etc.

Drugs

The substance used for the treatment of diseases and for reducing pain are called drugs. The process of treating disease by using drugs is called chemotherapy. **Paul Ehrlich** is called as **'Father of Chemotherapy'**.

Characteristics of drug

1. The action of the drug should be localized at the site where it is desired to act.
2. It should be non-toxic.
3. It should have minimum side effect.
4. It should not injure host tissue.
5. It should be efficient.

Classification of drugs on the basis of origin

1. Natural drugs

Those drugs which are obtained and extracted from plants and animals are called natural drugs. Eg: jatamasi, cocaine, leaf of neem, garlic, coriander, etc. They have valuable medicinal value and have been used against various diseases. Penicillin is one of the oldest natural drugs.

2. Synthetic drugs

Those drugs which are synthesized artificially in labs or industries by different chemical reactions are called artificial drugs. Eg: paracetamol, aspirin, novalgin, melubrin, etc. they might be in the form of liquids, tablets, capsules, etc. they are more specific and effective against diseases. They are collectively called allopathic.

Classification based on therapeutic actions

1. Chemotherapeutic drugs

Those drugs which are used to treat infectious disease caused by microorganism are called chemotherapeutic drugs. They are used for treating bacterial, viral, protozoal, malarial infections, etc. Eg:

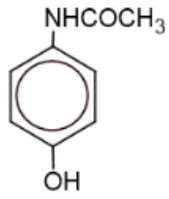
2. Pharmacodynamics drugs

Those drugs which are used to treat diseases caused by the functional disorder or degradation of the body, not by microorganisms are called pharmacodynamics drugs. They are used for treating cancer, diabetes, depression, etc. Eg:

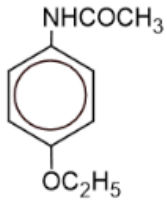
Classification on the basis action against various pathogens or psychological effects

1. Antipyretic drug

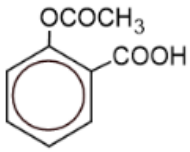
- To lower body temperature during high fever



Paracetamol



Phenacetin



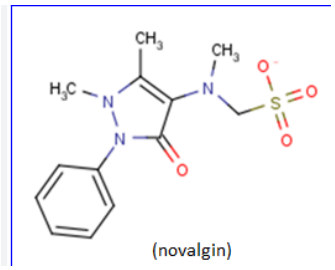
Aspirin

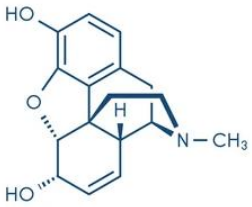
- Phenacetin and aspirin also relives pain i.e. analgesic

2. Analgesics

- To relieve pain without loss of consciousness
- Known as painkiller
- Two types:

a. **Narcotics (habit-forming):** induce sleeping as well as may cause unconsciousness if taken an overdose, responsible for drug addiction. Eg: morphine, heroin, etc.
b. **Non-narcotics (non-habit forming):** do not induce sleep and no loss of consciousness. Eg: Aspirin, novalgin, brufen, etc.

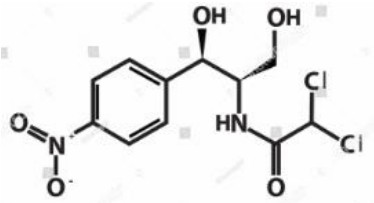

(novalgin)

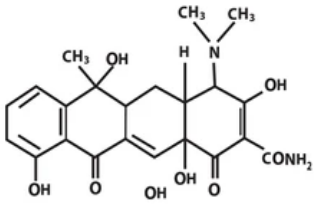

morphine

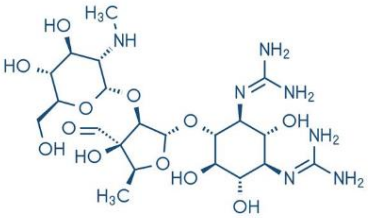
HEROIN

3. Antibiotics

- Derived from the chemical substance produced by microorganisms like bacteria, fungi, etc., and can be used to kill other microorganisms. Eg: penicillin, chloramphenicol, streptomycin, etc.
- Two types:
  a. **Broad spectrum**: effective against a wide range of microorganisms. Eg: chloramphenicol, tetracycline, etc.
  b. **Narrow spectrum**: effective against a few microorganisms. Eg: penicillin, streptomycin, etc.

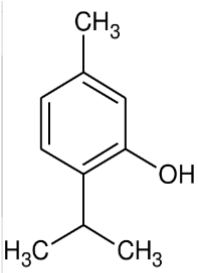

Chloramphenicol


Tetracycline

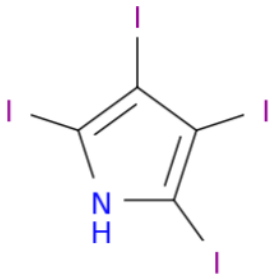

streptomycin

4. Antiseptic and disinfectant drugs

- To kill or prevent the growth of microorganisms in wounds, cuts
- Applied directly in the external portion of infected parts but disinfectants can't be applied to living tissues.
- Antiseptics: Dettol, thymol, iodol, etc.
- Disinfectants: chlorine water, phenol, SO_2, etc.



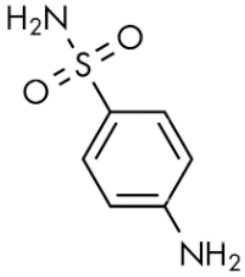
Thymol



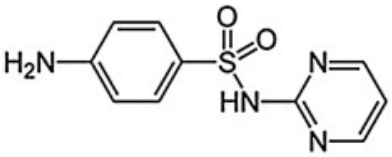
Iodol

5. Sulpha-drugs

- For the treatment of bacterial disease
- First chemotherapeutic agent which are derivatives of sulphanilamide
- Eg: sulphadiazine, sulphapyridine, sulphaguanidine, etc.



sulfanilamide



sulfadiazine

6. Anaesthetics

- Brings the loss of sensation for a certain period of time
- Uses made surgery comfortable for patients and allowed more time for doctors to work on care and more effectively
- Two types:
  a. **General or central anesthetics:** cause unconsciousness all over the body. Eg: diethyl ether, chloroform, nitrous oxide (N_2O), etc.
  b. **Local anaesthetics:** cause unconsciousness to a limited region. Eg: benzocaine, ethyl chloride, procaine, etc.

7. Tranquilizers (Hypnotic)

- Use for treatment of mental disorder and brings calmness to the respective person

- Act on the central nervous system and reduce anxiety
- Also called psychotherapeutic drugs
- Habit-forming drugs
- Eg: Equanil, dizepam, barbituric acid, etc.

8. **Cardiovascular drugs**

- For the treatment of heart and blood vessel diseases like hypertension, high blood pressure, etc.
- Eg: amyl nitrite, glycerol trinitrate, etc.

Drug addiction

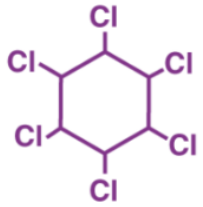
It is a compulsive desire and dependence on an illegal drug or a medication despite its consequences. Gene expression changes as the addiction progresses, affecting how the brain appears and works and, in turn, how behavior is shown. It encompasses extreme mood swings, anxiety, depression, and aggressiveness. It also covers the consequences of mental illness, confusion, hallucinations, and a loss of daily enjoyment. An individual's capacity for voluntary decision-making is hampered by these brain alterations, which result in compulsive drug yearning, seeking, and usage. It has a variety of negative physical, physiological, and psychological repercussions that have a significant negative impact on an individual's life, family, and society as a whole. Drug addiction is caused by numerous narcotic drugs, including morphine, heroin, cocaine, etc.

Pesticides

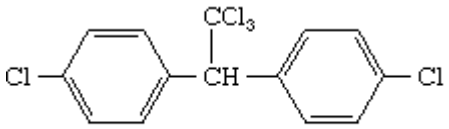
The substances that are used to kill harmful pests in agriculture are called pesticides. Eg: DDT, BHC, pentachlorophenol, etc.

Types of pesticides

1. **Insecticides:** used to kill or control insects, Eg: BHC (Benzene Hexa Chloride), DDT (Dichloro Diphenyl Trichloroethane), etc.

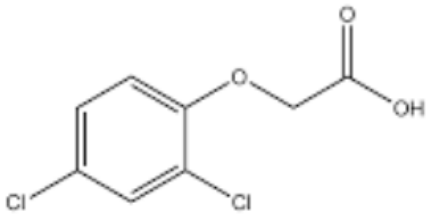


BHC



DDT

2. **Weedicides/ herbicides:** used for killing weeds or unnecessary herbs, Eg: 2,4-D (Dichlorophenoxyacetic acid), 2,4,5- T, etc.



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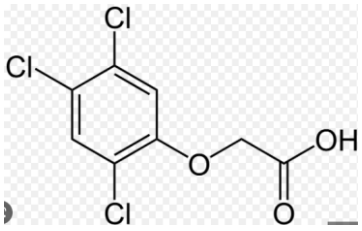
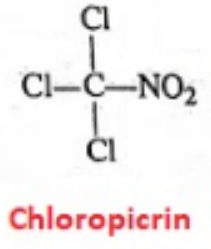
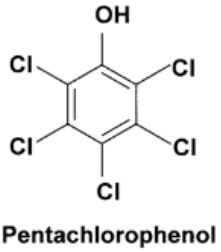


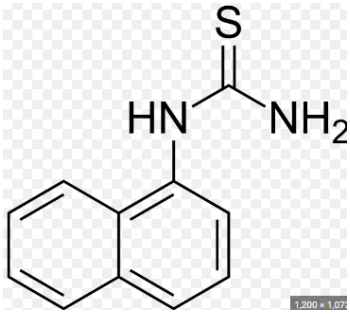
2, 4, 5 - T

3. **Fungicides:** used to kill or control fungi, Eg: $CuSO_4$, chloropicrin, aluminium phosphide, pentachlorophenol, etc. elemental sulphur is the oldest fungicides for grapes, strawberry, and many vegetables.





4. **Rodenticides:** used to kill and control rodents, Eg: sodium monochloroacetate, zinc phosphide, ANTU (Alpha napthyl thiourea),etc.



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